DSA0404 - FUNDAMENTALS OF DATA SCIENCE
UNIT 1 - ASSESSMENT TOOL 1 (CO1-AT1)
SHORT ANSWER QUESTIONS

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1. What is Data Science? Explain its primary objective.

Data Science is an interdisciplinary field that combines statistics, programming, domain expertise, and machine learning to extract meaningful insights from data.

Its primary objective is to convert raw data into actionable intelligence that supports informed decision-making in organizations and drives business value through predictive and prescriptive analytics.

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2. Why is Data Science important in today's digital world? Mention any four reasons.

→ Enables data-driven decision making for competitive advantage

→ Helps organizations identify patterns and trends in vast volumes of data

→ Improves operational efficiency and reduces costs through predictive insights

→ Supports innovation and development of new products and services

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3. List any four benefits of Data Science in business and society.

→ Improved business intelligence and strategic planning
→ Enhanced customer experience through personalized insights
→ Risk mitigation and fraud detection capabilities
→ Acceleration of scientific research and medical discoveries

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4. Mention four real-world applications of Data Science.

→ Healthcare: Disease prediction, treatment optimization, and drug discovery

→ Finance: Credit scoring, fraud detection, and algorithmic trading

→ Retail: Customer segmentation, recommendation systems, and demand forecasting

→ Transportation: Route optimization, autonomous vehicles, and traffic prediction

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5. What are the different facets of data? Briefly explain each.

The different facets of data are structured data, semi-structured data, and unstructured data.
→ Structured Data: Organized in rows and columns (Ex: relational databases).
→ Semi-Structured Data: Has a flexible structure using tags or keys (Ex: XML, JSON).
→ Unstructured Data: Has no predefined format (Ex: Social media posts).

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6. Differentiate between structured, semi-structured, and unstructured data with suitable examples.

Structured Data	Semi-Structured Data	Unstructured Data
Organized in predefined formats with clear schema	Has some organizational properties but lacks rigid schema	No predefined format or organization
Stored in relational databases	Includes metadata for context	Requires processing to extract meaning
Ex: Customer records	Ex: JSON, XML, emails	Ex: Videos

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7. What is Big Data? State its key characteristics (5 Vs).

Big Data refers to extremely large datasets that require specialized tools and techniques for processing.

The 5 Vs are:

→ Volume
→ Velocity
→ Variety
→ Veracity
→ Value

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8. Explain the Big Data ecosystem. Name any four technologies used in it.

The Big Data ecosystem comprises tools, frameworks, and platforms for collecting, storing, processing, and analyzing large datasets.

Four key technologies:

→ Hadoop
→ Spark,
→ HBase
→ Kafka

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9. What are the major phases of the Data Science process?

The major phases of the Data Science process are:

→ Problem definition and business understanding
→ Data collection and acquisition
→ Data cleaning and preprocessing
→ Exploratory data analysis (EDA)
→ Feature engineering and selection

- → Model building and training
- → Model evaluation and validation
- → Deployment and monitoring

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10. What is data retrieval? Mention two common sources from which data can be retrieved.

Data retrieval is the process of accessing and extracting data from various sources for analysis and processing.

Two common sources:

- → Databases
- → APIs

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11. Why is data collection considered an important step in the Data Science process?

- → Data collection is crucial because the quality and quantity of data directly impact model performance and insights.
- → Accurate data collection ensures relevance to the problem, reduces bias, enables comprehensive analysis, and forms the foundation for all subsequent phases.
- → Poor data collection leads to unreliable models and misleading conclusions.

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12. What is data cleaning? List any four common data-cleaning techniques.

Data cleaning is the process of identifying and correcting errors, inconsistencies, and missing values to improve data quality.

Four techniques:

- → Handling missing values
- → Removing duplicates
- → Correcting data type errors
- → Treating outliers and anomalies.

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13. What is data integration? Why is it necessary in Data Science?

Data integration is the process of combining data from multiple, heterogeneous sources into a unified view.

Integration ensures data consistency, eliminates silos, provides a complete picture of information, and enables comprehensive analysis and accurate insights.

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14. Explain data transformation with suitable examples.

Data transformation is the process of converting data from one format or structure to another for analysis.

Examples: Normalization, Aggregation, Encoding, Pivoting, Binning

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15. What is data analysis? Mention any four techniques used for data analysis.

Data analysis is the systematic examination of data to discover patterns, relationships, and insights.

Four techniques:
- → Descriptive statistics
- → Regression analysis
- → Clustering
- → Classification

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16. Differentiate between descriptive, predictive, and prescriptive analytics.

Descriptive Analytics	Predictive Analytics	Prescriptive Analytics
Analyzes historical data to understand what happened	Uses statistical models & ML to forecast future outcomes	Recommends actions based on predictions to optimize outcomes
Ex: Sales trends	Ex: Customer churn prediction	Ex: Inventory optimization strategies

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17. What is meant by building a model in Data Science? Mention the basic steps involved.

Building a model involves creating a mathematical representation of data patterns to make predictions.

Basic steps:
- → Select appropriate algorithm based on problem type
- → Split data into training and testing sets
- → Train the model on training data
- → Validate and tune hyperparameters
- → Evaluate performance on test data
- → Finalize and prepare for deployment.

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18. Why is model evaluation important? Name any four evaluation metrics used for classification models.

Model evaluation assesses model performance, identify generalization, and ensure reliability before deployment.

Four metrics:
- → Accuracy
- → Precision
- → Recall
- → F1-Score

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19. Why is presenting findings an essential part of the Data Science process? Mention any four visualization tools or techniques.

Presenting findings communicates insights to stakeholders, facilitates decision-making, and ensures insights are actionable.

Four visualization tools/techniques:
- → Bar charts
- → Line graphs
- → Scatter plots
- → Heatmaps

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20. How are Data Science models deployed into real-world applications? Give any four examples of Data Science applications.

Models are deployed through containerization, cloud platforms, APIs for integration, or embedded directly into applications with continuous monitoring.

Four applications:
- → Recommendation systems
- → Fraud detection
- → Predictive maintenance
- → Natural Language Processing
